

Paper V — The Untrusted Planner

Keep Flying Whatever the Model Says: Predictive Safety Filtering for Language-Model UAV Agents under Adaptive Attack

Research proposal from the RobustUAVs.ai programme · Roger Nick Anaedevha · 2026-10-01
Target: IEEE SaTML (fallback: USENIX Security) · builds on the parent manuscript [1] and Paper III

Abstract

Language-model agents are beginning to sit in the high-level loop of autonomous aircraft: translating an operator’s instruction into waypoints, summarising telemetry, choosing a task, proposing a mode change. That places an attackable, hard-to-secure component upstream of a safety-critical controller, and the natural question — can we stop the language model from being fooled? — has no general answer. We propose to stop asking it. A *predictive safety filter* sits between the agent’s proposed action and the actuators: it admits the action only if the certified navigation model (the parent manuscript’s) has a recursively feasible trajectory keeping the aircraft inside its safe envelope over the planning horizon, and otherwise overrides to a verified fallback. The security claim is then independent of the language model’s internals: for *any* sequence of agent outputs, adversarial or not, the physical state stays in the certified safe set. An adaptive attacker who fully controls the agent’s inputs is thereby downgraded from causing a crash to causing a *stall* — the filter converts a safety attack into a liveness attack, and we quantify the residual it cannot prevent. We make three contributions: the safety-invariance theorem for an untrusted planner; an adaptive-attack evaluation protocol, following the discipline that a defence must be tested against an adversary who knows the filter; and the measured trade between the filter’s conservatism and the mission-denial rate an attacker can force, run on the RobustUAVs.ai simulator with real language-model agents in the loop.

1 The problem, and the move

Autonomy stacks increasingly place a language-model agent in the planning loop [6]: it reads a textual description of the situation — operator messages, telemetry summaries, a map, peer reports — and emits a high-level action. That channel is attackable. Indirect prompt injection [3] lets an adversary who controls any text the agent ingests — a spoofed operator note, a tampered telemetry string, a poisoned map annotation — steer the agent’s output. Securing the language model against this *in general* is, on current evidence, not something anyone can promise.

The move this paper makes is to stop trying. Borrowing the logic of the Simplex architecture [7, 8] and the predictive safety filter [10, 4], we treat the agent as an *untrusted high-level planner* and interpose a filter that trusts only the certified navigation model:

The language model may propose anything. The filter executes the proposal only when a certified safe trajectory exists that begins with it; otherwise it executes a verified fallback. Physical safety then holds for every possible agent output, so whether the attacker fooled the model becomes irrelevant to whether the aircraft stays in its envelope.

This is the same object as Paper III’s projection layer, with two changes that make it a security paper rather than a control one: the source of the proposed action is an *adversarially controllable* language model rather than a trained policy, and the evaluation is against an *adaptive* attacker who knows the filter.

2 Where MPC and the language model meet

The filter is a model-predictive control problem [5]. Let the navigation state evolve as $\dot{x} = F(x, u)$ (F Lipschitz with constant L over the operating region, as in the parent manuscript), let $\mathcal{S}$ be the mission’s safe set (corridor margin m , certified radius $R_i(t) \geq R_{\min}$), and let the language model propose a high-level action a_t^{LLM} (a waypoint, a task, a mode).

Definition 1 (Predictive safety filter). *Given a_t^{LLM} , solve*

$$a_t^* = \arg \min_a \|a - a_t^{\text{LLM}}\|$$

subject to: a certified trajectory from x_t under a stays in $\mathcal{S}$ over the horizon H and ends in a control-invariant terminal set. Execute a_t^ if the problem is feasible; else execute the verified fallback (loiter, or return-to-launch).*

The admissible set $\mathcal{A}_{\text{adm}}(x_t)$ is exactly the certified set the parent paper’s budget produces — the filter is where the certificate is *used*, not just reported. When $a_t^{\text{LLM}} \in \mathcal{A}_{\text{adm}}$, the agent’s action passes untouched; the filter is transparent in the benign case and binds only when the proposal would leave the envelope.

3 Three results

3.1 Safety is invariant to the planner (the headline)

Proposition 1 (Safety under an arbitrary planner). *If the terminal set is control-invariant and the fallback is recursively feasible from it, then under the filter the physical state satisfies $x_t \in \mathcal{S}$ for all t , for every sequence $\{a_t^{\text{LLM}}\}$ — including one chosen by an adversary with full control of the agent’s inputs and full knowledge of the filter.*

This is recursive feasibility of the predictive safety filter [10] carried to an adversarial planner. The content is not the MPC machinery, which is standard, but the security reading: the attacker’s control of the language model does not appear in the conclusion. The safety guarantee has been made *independent of the component that cannot be secured*.

3.2 The attack surface collapses to liveness

Proposition 2 (Safety attack $\rightarrow$ liveness attack). *Under the filter, an adaptive adversary controlling the agent’s inputs cannot cause a safety violation (Prop. 1); its maximal effect is to force the filter to override, i.e. to induce the fallback. Its power is therefore bounded by the induced fallback rate — mission denial — and never extends to envelope violation.*

This reframing is the paper’s point. “Can the attacker crash the drone?” (no, under the filter) is replaced by “how often can the attacker stall the mission?” (a measurable rate), which is both the right security question and one the platform can answer.

3.3 The language attack and the physical attack share one budget

Proposition 3 (Composition with the perturbation budget). *A concurrent navigation-degradation attack (jamming, spoofing) shrinks $\mathcal{A}_{\text{adm}}(x_t)$ through the same budget δ_{tot} of Paper IV. The induced fallback rate is therefore monotone in δ_{tot} : the language-channel attack and the physical-channel attack compose, and the filter’s conservatism is the common currency.*

This is the bridge to the rest of the programme: Paper V’s attacker and Paper IV’s attacker draw on one certified budget, exactly as Paper I’s credit/debit did.

■ *Proposition 1 guarantees safety, not liveness. The filter keeps the aircraft in its envelope; it does not make the mission succeed, and a sufficiently aggressive attacker can hold the system in fallback indefinitely (persistent denial). That is a real limitation and the honest headline is the pair: safety is unconditional under the stated model, liveness is not. The paper must report the denial rate as prominently as the zero-violation result, or it oversells.*

4 Threat model

- **Adversary capability.** Full control of any text the agent ingests (indirect prompt injection [3]); full knowledge of the filter, the certified model, and the fallback. Optionally, a concurrent bounded navigation-degradation attack (Paper IV’s budget).
- **Adversary goal, in order of severity.** (i) A physical safety violation — leave the corridor, breach separation. (ii) Failing that, mission denial — maximise the fallback rate. (iii) A *stealthy* denial that the operator does not attribute to an attack.
- **Trust boundary.** The language model is untrusted. The certified navigation model, the filter solver, and the fallback are trusted (and are small enough to be verified, which the language model is not). The paper is explicit that it secures the *system*, not the model.
- **Out of scope.** Attacks on the trusted base (the filter solver, the certified model itself) — these are the assumptions, stated as assumptions.

5 Adaptive evaluation (the methodological core)

A safety filter evaluated against a *fixed* attack proves nothing: the lesson of a decade of adversarial-ML evaluation [2, 9] is that a defence must be tested against an adversary adaptive to it. We carry that discipline to the physical setting.

RQ1 Does the filter hold the zero-violation guarantee empirically, under an adaptive adversary?

Expected: zero physical violations, matching Prop. 1; any violation is a bug in the instantiation (an infeasible solve executed, a model-mismatch term exceeded) and is treated as one.

RQ2 What denial rate can an adaptive adversary force, as a function of its control over the input channel and its knowledge of the filter?

RQ3 How does denial scale when the language attack composes with a navigation-degradation attack (Prop. 3)?

RQ4 What does the filter cost when there is no attack? The benign-case intervention rate and mission-time overhead — the conservatism price of the guarantee.

Adaptivity, concretely and defensively. The adaptive adversary is constructed to probe the filter’s *boundary* — the inputs most likely to drive the agent toward the edge of $\mathcal{A}_{\text{adm}}$ — because that is what a sound evaluation of the *defence* requires. The deliverable is the protocol and the measured denial curve, not an attack toolkit: following the parent manuscript’s handling of its evasion analysis, the mechanism is described at the level needed to reproduce the evaluation, the result *bounds* what the attacker achieves, and the practical output is a filter-configuration procedure.

6 What is in hand, what is new

In hand	To build
Certificate engine producing $\mathcal{A}_{\text{adm}}$ (parent; $\gamma = 1.625$ m/s supremum)	the MPC safety-filter solver over $\mathcal{A}_{\text{adm}}$ with a verified fallback
Fleet simulator (MAX_FLEET=12, $T = 70$ s, $\kappa = 0.25$) and ten attack classes	a language-model agent in the planning loop (the platform already integrates anthropic / openai / gemini / deepseek)
Paper IV’s perturbation budget δ_{tot}	the adaptive-attack protocol and the denial-rate measurement
Paper III’s projection/feasibility machinery	its lift from discrete allocation to continuous MPC under an untrusted planner

7 Experimental plan on RobustUAVs.ai

1. **Agent in the loop.** Put a language-model planner (the platform’s existing providers) on the mission-planning task of the fleet simulator: it proposes waypoints, task assignments and mode changes from a textual situation description.
2. **Filter.** Implement the predictive safety filter using the certificate engine for $\mathcal{A}_{\text{adm}}$ and a loiter/RTL fallback; log every intervention.
3. **Adaptive attack.** An adversary controls the textual channel feeding the agent and adapts to the filter. Measure physical violations (expect zero), the induced fallback rate, and stealthy-denial success.
4. **Composition.** Repeat under a concurrent navigation-degradation attack from Paper IV’s budget, to trace the denial surface over δ_{tot} (Prop. 3).
5. **Benign cost.** Intervention rate and mission-time overhead with no attacker (RQ4).
6. **Platform page.** A *Safety Filter* page showing the agent’s proposal, the filtered action, the admissible set, and the live fallback rate under attack — reusing the existing export and provenance machinery.

■ *Model-plant mismatch is the quiet assumption. Prop. 1 holds for the certified model; a gap between it and the simulated dynamics must be carried as an explicit robustness margin in the filter; and on real hardware it would have to be identified, not assumed. The simulation study can state the margin it uses; it cannot claim hardware safety, and must not.*

8 Why this venue, and why it is distinct

IEEE SaTML is the fit: secure and trustworthy machine learning, with language-model agent security and adaptive-attack evaluation squarely in scope, and a control-theoretic safety mechanism as the defence. USENIX Security is the fallback if the systems/benchmark side dominates.

Distinct from the others in object:

- **Paper III** filters the output of a *trusted* RL policy for feasibility; Paper V filters the output of an *untrusted* language model for safety, and is evaluated adversarially.
- **Paper IV** attacks the *sensing* channel; Paper V attacks the *planning* channel, and Prop. 3 shows they share a budget.
- **Paper II** would classify Paper V as a runtime-assurance lift — a per-operation guarantee discharged by a monitor rather than an offline certificate — which is exactly the bridge Paper II names for objectives no static certificate can reach.

9 Risks

- **“The guarantee is just MPC recursive feasibility.”** True, and stated: the novelty is the security reading (safety independent of an unsecurable component), the adaptive-attack evaluation, and the composition with the sensing budget — not the MPC theorem.
- **Denial might be trivially easy.** If any adaptive attack forces near-100% fallback, the filter guarantees safety but no useful liveness, and the honest finding becomes “safety filtering is necessary but not sufficient; the language channel needs its own defence”. That is still a result, and it motivates the next paper.
- **Adaptive-attack completeness.** We cannot prove we found the strongest attack; we report the protocol and the achieved denial, per the discipline [9], and never claim a worst-case denial bound we did not prove.

- **Dual-use.** The adaptive-attack construction is dual-use. Following the parent manuscript, it is presented with its mitigation (the filter) in the same work, bounds what the attacker achieves, targets a defensive artifact, and is evaluated in simulation against the project’s own defensive platform. An ethics section is mandatory.

10 Page plan and timeline

Section	pp	Section	pp
Introduction	1.0	Adaptive evaluation + results	2.5
Threat model	1.0	Composition with sensing	1.0
Filter and MPC formulation	1.5	Discussion, ethics, limits	1.0
Safety-invariance theory	1.75	Related work, conclusion	1.0
Body (SaTML / USENIX range)			10.75

About five months. The certificate engine, the fleet simulator and the language-model integration already exist; the new engineering is the MPC filter, the agent-in-the-loop harness and the adaptive-attack protocol. The safety theory is standard MPC recursive feasibility, so it can be written early; the risk and the interest are both empirical — how much liveness the filter costs, and how much an adaptive attacker can take away.

References

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